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Pseudocode Manual

for the IRSIA-FNRS Computer

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P r é f a c e

Le code de la calculatrice IRSIA-FNRS, décrit dans le manuel de programmation, est assez lourd par suite de la flexibilité des préfixes et de la nécessité du groupement des instructions en paires. La programmation est facilitée si l'on fait usage d'un système de notations plus courtes et plus simples, que nous appellerons pseudocode, que la machine est capable de traduire en son code régulier grâce à une routine de traduction élaborée une fois pour toutes.

Le présent manuel décrit le pseudocode (chap. I) et donne l'information nécessaire pour permettre à un opérateur d'exploiter la machine pour effectuer la traduction d'un problème rédigé en pseudocode (chap. II). La lecture du chap. III n'est pas nécessaire pour l'exploitation : le programme de traduction y est consigné et expliqué.

Le pseudocode et le programme de traduction ont été mis au point par R. Gould, qui a rédigé le présent manuel directement en anglais, tout en gardant la terminologie française du manuel de programmation pour les désignations spéciales.

Le chap. IV décrit quelques routines assez courtes mémorisées sur le même ruban que les programmes de traduction. Il se termine par un appendice, rédigé en français par A. Fischer, et un second appendice en anglais par R. Gould, apportant quelques corrections et suppléments au manuel de programmation.

III

Table of Contents

Chapter I : Use of the Pseudocode

1. List of Orders	1
1.1. Pseudo-orders	
1.2. Other special addresses	
1.3. The index digit	
2. Pseudocode Programming	4
2.1. Recording pseudoprograms	
2.2. Translation	
3. Description of Special Pseudo-orders	7
3.1. Prefix 38	
3.2. Prefix 39	
3.3. Prefix 05 and renvois	
3.4. Prefixes 70-89	
4. Use of the Eighth Digit	10
4.1. Tranches and groups	
4.2. Operations on orders	
4.3. Renvoi designators	
5. An Example : "Programme de transfert et de contrôle"	12

Chapter II : Use of the Translation Routine

1. General Information	15
1.1. Reading of the routine	
1.2. Pupitre d'impression settings	
1.3. Pupitre de commande settings	
1.4. Starting the routine	
2. Special Printouts	18
2.1. Type 1 printouts	
2.2. Type 2 printouts	
3. Stops	19
3.1. Stops without alarm	
3.2. Stops with alarm	
4. Modifications	20
4.1. Prefixes 71-89	
4.2. Storage requirements	

Chapter III : Details of the Translation Routine

1. Introduction	22
2. Working Storage	22
2.1. Part I	
2.2. Part II	
3. "Programme de Transfert et de Contrôle", 0015-0090	22
4. Block α , 0095-0675	25
5. Block β , 0700-1115	28
6. Block 7, 1130-1205	31
7. Block 5, 1210-1270	31
8. Block 4, 1280-1515	32
9. Block ϵ , 1520-1650	34
10. Block 1, 1660-1680	35
11. Block 0, 1730-1865	35
12. Block δ , 1900-2030	36
13. Block ρ , 2070-2600	37
14. Constants	40
15. Block Diagrams	43

Chapter IV : Other Routines

1. Pseudocode Printing Routine	52
1.1. Use of the routine	
1.2. Details of the routine	
2. Read-Off-Drum Routine	54
2.1. Use of the routine	
2.2. Details of the routine	

<u>Appendice I</u> : Programme de transfert et de contrôle	57
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<u>Appendice II</u> : Addenda to the "Manuel de Programmation"	59
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Chapter I : Use of the Pseudocode

1. List of Orders

1.1. Pseudo-orders

ORGANISATION

90 org. tambours
94 a' = S
95 org. transfert
99 org. rubans
60 virgule flottante
65 virgule fixe
36 bloc. al.
37 débloc. al.
38 eff. al. (but see § 3.1)
39 specify address of next order word

TRANSFERT

50 A → impr.
51 A → E
52 A → F
53 A → C
54 A → tamb.
(augment by 05 for B)

GROUPE CALCULATEUR

01 +	06 +*
02 -	07 -*
03 x	08 ⊗
04 x-	09 ⊗-

(augment by 10 to use the address as a number)

10 =	15 →
20 = impr.	25 → impr.
30 = C	35 → C

RENVOI ou FIN

05 designate next order word for a renvoi
40 Rv
45 Rv si - dans g. c. et vidange du g. c.
41 Rv si - dans ch.
46 Rv ex. al.
42 fin
47 fin al.
43 Rv inh. V=S
48 Rv inh. f. t.
44 Rv ex. C₁
49 Rv ex. C₂

DEROULEMENT

91 A, jusque fin groupe
92 A, jusque fin ruban
93 A, jusque a'
(augment by 05 for B)

ADRESSES

<u>lect.</u>		<u>inscr.</u>
E	0001	E
F	0002	F
A sans p. a.	0003	(vidange du g.o.)
B sans p. a.	0004	ch
V	0006	
W	0007	W
A avec p. a.	0008	
B avec p. a.	0009	

ORDER 99, "ORG. RUBANS"

Digits x, y, z select the tapes for positions A, B, C respectively.

IMPRIMEUR

Digit x controls the number of carriage returns made after the word is printed, as follows :

x = 0	0 returns
= 5	1 return
= 4	2 returns
= 2	10 returns

Digit y controls the printing of the vertical-line symbols, as follows :

		avant le mot	
		rien	
après	rien	0	5
le		2	7
mot		4	9

The word contained in E (with orders 20,25) or on the piste auxiliaire (with orders 50,55) also controls the printing. If the sign of this control word is +, the printer spaces twice after the modulus and then prints the sign of the exponent. If the sign is -, the spaces and the sign are omitted. Each of the 17 digits of the control word affects the printing of the corresponding digit of the printed word as follows :

0	impression normale
1	impression avec espace avant
2	impression avec point décimal avant
4	suppression

RUBANS

Ordinary addresses of words on tape end in 0 or 5. They are, however, modified as follows :

to signal:

increase address by :

fin de tranche
fin de groupe
fin de tranche et de groupe

1
4
3

The first word on each tape is a blank with the address 0002, "fin de ruban".

1.3. The index digit

The effect of the index digit "i" is shown in the following tables.

MOST ORDERS

i = 0
= 1-4
= 5
= 6-9

no effect
augment address by Z_1
augment address by M
augment address by $Z_1 - 5 + M$

ORDERS 50, 53-55, 58, 59

i = 0
= 1-4

no effect
augment each address read from
tape by Z_1

ORDERS 61-64, 67-69

i = 1-4

identifies the register Z_1 affected

2. Pseudocode Programming

A single pseudo-order is made up of a two-digit prefix, possibly followed by a four-digit address and an index digit in the following fashion :

xx	xxxx	x
prefix	address "a"	index "i"

It is suggested that the index not be written unless it is non-zero, and that the address not be written unless it is pertinent or is followed by a non-zero index. (However, non-pertinent addresses and index digits are not altered by the translation process; for example, the pseudo-order "42 8888 3" is translated as "42300 8888".) The address is not pertinent in the case of the following prefixes : 00(vacuous order), 21-24, 26-29, 31-34, 36-38, 42, 47, 50-60, 64, 65, 69, 71-78, 91, 92, 97. The index is not pertinent in the case of the follo-

wing prefixes : 00, 05, 21-24, 26-29, 31-34, 36-39, 42, 47, 51, 52, 56, 57, 60, 65, 66, 71-78, 91, 92, 96, 97; nor with 01-04, 06-10, 15 when those prefixes are followed by addresses 0001-0004, 0006-0009. The index is limited to less than 5 in the case of the following prefixes : 50, 53-55, 58, 59, 61-64, 67-69.

The pseudo-orders are not numbered or arranged in pairs. Renvois are made through the use of the special code 05, as explained in § 3.3.

Most of the pseudo-orders are in obvious one-to-one correspondence with regular machine orders. The exceptions are discussed in § 3. In most cases, the address written with a pseudo-order is the same as that written with the corresponding regular order. The only exceptions to this are the renvoi orders (discussed in § 3.3), and orders for transferring numbers between the groupe calculeur and E, F, V, W, A, B. In the regular code, such a transfer is made using a special prefix and a vacuous address; in the pseudocode, it is made using an ordinary prefix and a special address (0001, 0002, etc) designating the register or tape. These addresses are distinguished from ordinary addresses in that the latter always end in 0 or 5.

Orders for operations on orders (+OR1, -OR1, and so on) can be written in pseudocode by a method described in § 4.2. The pseudocode has no provision, however, for writing orders for multiplication without rounding or for high precision fixed-point calculations.

2.1. Recording pseudoprograms

Each pseudo-order is recorded as the first seven digits of an ordinary number word. Normally, the first pseudo-order on a tape will be preceded by a blank word having serial number 0002 (fin de ruban). The serial numbers recorded with the pseudo-orders are entirely arbitrary. It is recommended that they be arranged to increase automatically by 0005 (as described in the Manuel de Programmation, p. 60), for ease in making corrections. Note that additional pseudo-orders can later be inserted in the sequence without any need for re-numbering the words.

All of the pseudo-orders for a given program should be recorded in a single tranche. Whatever constants are required for the program should then be recorded on the same tape, in one or more tranches. If for some reason it is desired to generate a program which is divided into two or more tranches or groups, this can be done by a method described in §4.1. However, it is not possible to produce directly a tape on which tranches of orders are interspersed with tranches of numbers.

As soon as a pseudoprogram, followed by its accompanying constants, has been completely recorded on a tape, it should be printed out for verification. The routine described in Chapter IV, §1, is convenient for printing pseudoprograms.

2.2. Translation

Once the pseudoprogram has been verified, it is translated into a regular machine program by a translation routine which is described in Chapter III and III. The routine translates each individual order into regular machine code, arranges the orders in pairs (inserting vacuous orders when necessary), numbers the pairs in order, and records the translated program on a second tape. Ordinarily, the first order-word of the translated program has serial number 0015, and the last is accompanied by a "fin de tranche et de groupe" indication. However, either of these arrangements can be modified if desired (see §3.2, §4.1).

The translation routine initially records a blank word with serial 0002. After the translation is completed, the routine copies unchanged the remaining contents of the original tape (i. e., everything after the first tranche) onto the new tape. Either of these features can be suppressed if desired.

While it is translating the pseudoprogram, the translation routine may, if desired, print out a partially translated program, referred to here as a "semi-pseudoprogram" or simply "semiprogram". In this printout the orders are organized in pairs and numbered as in the final program. In fact, the printout is identical to that of a regular program except that each regular order is

written as the corresponding pseudo-order. (Thus the special pseudoprefixes 05, 39, 45, 70-89 do not appear.) In some cases, this "semiprogram" printout may prove to be sufficiently useful to make a printout of the regular program unnecessary. However, its printing slows the translation routine considerably.

3. Description of Special Pseudo-orders

3.1. Prefix 38

Normally 38 is the pseudoprefix for "effacement alarme", corresponding to regular prefix 00034. However, if 38 appears in the first pseudo-order on the tape, it is not interpreted as "effacement alarme"; instead, it serves as a signal for the translation routine to begin the generated program with a "programme de transfert et de contrôle". The orders of this program are numbered from 0015 to 0090, and so the next order in the generated program will be numbered 0095.

The effects of this "programme de transfert et de contrôle" are as follows : the second tranche on the tape is transferred onto the number drum; the contents of the first tranche is checked, word by word, against the contents of the order drum; the contents of the second tranche is similarly checked against the number drum; and then there is a renvoi to order 0095. The contents of both first and second tranche may be in any order, i. e., the words need not be numbered consecutively. The complete coding of this program is given in § 5.

3.2. Prefix 39

As noted above, the orders of the translated program are normally numbered consecutively, starting with 0015. Prefix 39 is used if a break in this scheme of numbering is desired. The effect of the pseudo-order "39 xyzu" is to number the next translated order-word "xyzu", and to number the following order-words consecutively. Such an order may appear as the first one in the pseudo program, if it is desired to begin the translated program at some address other than 0015. The last digit, "u", of the address accompanying prefix 39 should always be 0 or 5.

It is assumed that order 39 will be used infrequently. If it is desired at some point in a program, it will ordinarily fix 00 as many times as desired.

and not plan to use location 0000 on the order translation routine causes vacuous orders to

40 1000 = renvoi au 6^{ème} drum
de l'adresse 1000

ou xyzu

05 xyzy or more

pointum
dites no:
6^{ème} drum

not numbered, renvoi instructions are not at each point in the program to which a renvoi is made. This "Designator" is an arbitrary 4-digit number. This "Designator" is the last section of every renvoi order referring to that point. The pseudo-order "40 1000" does not mean "renvoi to location 1000", but rather "renvoi to the point in the program which has been assigned the designator 1000". Prefix 05 is used for assigning the designators. Thus, for each designator xyzu, the pseudo-order "05 xyzu" is inserted at the point in the program to which the renvois are to be made. (The next order following "05 xyzu" is the first one to be performed after a renvoi "40 xyzu".) When the pseudo program is translated, the arbitrary designators are replaced by the appropriate addresses of order-words on the order drum.

The following facts about the assignment of renvoi designators should be noted: Any 4-digit number, 0001-9999, may be used as a designator. The designators need not appear in numerical order in the program. The total number of designators used in any one program may not exceed 880; hence for practical purposes the number may be regarded as unlimited. More than one designator may be assigned to any point in the program, i.e., two or more "05" orders may occur consecutively. Care should be taken to see that the same designator is not assigned to more than one point, i.e., there must not be two "05" orders with the same address. However, if this should be done inadvertently, the translation routine will give an alarm. If a "39" order appears in the pseudo-program,

an adjacent "05" order should follow it rather than precede it; otherwise a renvoi order referring to that renvoi point will not be translated correctly.

The designator 0000 is always interpreted as designating the address 0000. Hence, for example, the pseudo-order "41 0000 4" is translated as "41400 0000". The pseudo-order "05 0000" has no effect except to ensure that the following pseudo-order will be translated as the first order of an order-word.

A special feature of the translation routine is associated with renvoi orders accompanied by an index digit of 5. For example, the pseudo-order "40 1000 5" is treated as follows: the address of the translated pseudo-order is that corresponding to the designator 1000, diminished by $(L + 5)$, where L is the number of the order-word in which the translated order appears. It may be seen that this adjustment is necessary in order to make the resulting relative renvoi correct.

Code 45 is a special prefix. The pseudo-order "45 xyzu 1" is equivalent to the pair "10 0004, 41 xyzu 1". Hence it is translated into the two regular orders "07446 0000, 41100 XYZU", where XYZU is the address corresponding to the designator xyzu.

For some additional techniques involving renvoi orders and designators, see § 4.3.

On the "semiprogram" printed out by the translation routine, each designator is shown at the right of the order-word to which it corresponds. If more than one designator is assigned to the same point in the program, only the last is shown.

3.4. Préfixes 70-89

Prefixes 71-78 are used for renvois to the elementary function subroutines. For example, the pseudo-order "71 0000" is translated as the order word "01900 0000, 40000 8265". The address and index of these pseudo-orders are not pertinent.

Semiprogram

0015	90 0006	54 0000	
0020	92 0000	61 0000 3	
0025	95 0009	90 0005	
0030	52 0000	61 0001 4	1000
0035	48 0040	61 0000 4	
0040	06 0006	62 0000 1	2000
0045	0110000 1	0210002 0	
0050	0120000 1	0220002 0	
0055	10 0001	28 0000	
0060	01 0001	10 0004	
0065	41 0090	17 0000 4	
0070	10 0004	41 0030	
0075	17 0000 3	10 0004	
0080	41 0095	61 0001 3	
0085	90 0006	40 0030	
0090	42 8888	40 0030	3000

Program

0015	90000	0006	54000	0000
0020	92000	0000	06800	0000
0025	95000	0009	90000	0005
0030	52000	0000	06900	0001
0035	48000	0040	06900	0000
0040	02566	0000	06646	0000
0045	04168	0000	02078	0000
0050	04169	0000	02079	0000
0055	07146	0000	00082	0000
0060	02116	0000	07446	0000
0065	41000	0090	03476	0000
0070	07446	0000	41000	0030
0075	03376	0000	07446	0000
0080	41000	0095	06800	0001
0085	90000	0006	40000	0030
0090	42000	8888	40000	0030

Chapter II : Use of the Translation Routine

1. General Information

The IRSIA-FMRS Computer Pseudocode Translation (PCT) Routine is an ordinary machine program. A single tranche of pseudo-orders on one magnetic tape forms the input, and a corresponding sequence of regular orders is recorded onto another tape as the output. The translation routine consists of two parts. In Part I, the input tape is read and a table of renvoi designators, with their corresponding addresses in the translated program, is formed on the number drum. In Part II, the input tape is read again, the actual translation is made, and the output tape is produced.

The approximate average operation times for the translation routine are as follows :

Part I : .7 seconds per pseudo-order

Part II : 4.8 seconds per pseudo-order if semiprogram is printed out

1.6 seconds per pseudo-order without semiprogram printout

The actual time required will, of course, depend on the particular sequence of pseudo-orders. In particular, the time is quite dependent on the number of designators used; the above figures were obtained with a pseudoprogram containing 28 designators. No allowance is made in the above figures for special printouts or for tape copying, etc., at the end of the routine.

The words on the translation routine tape have the following serial numbers, and thus occupy the corresponding locations on the drums :

orders	0015 - 0090	"Programme de transfert et de contrôle"
	0095 - 2600	Translation routine
	7940 - 8168	Other routines (see Chapter IV)
numbers	0100 - 0608	Constants

Locations 0000-0095 on the number drum are used for working storage. The table of renvoi designators begins at location 0610, with the designators themselves occupying 0610, 0620, and so on.

1.1. Readin of the routine

If the two order-words :

0005	99000	ABCO	92000	0000
0010	54000	0000	99000	ABCO

are stored on the order drum, then it is necessary only to synchronize and to start the computer at order 0005. The PCT tape must be in position A. Alternatively, the orders drawn can be performed manually and the computer started at 0015. The "programme de transfert et de contrôle", which is placed in locations 0015-0090, is described in § 3 of the next chapter.

After the contents of the PCT tape have been transferred to the drums, each word is individually checked. If any word fails to check, there is a stop at order 0095 and the address "8888" may be seen in register N. The last word read from tape is in F, and its address is in Z_1 . In order to check the remaining words, one need only press "Programme Start" (except that if $Z_1 = 8168$, one must first set Z_3 to 1 and "Org. Tambours" to 6). When the checking is completed, the machine stops at order 0095.

The readin of the PCT routine requires about two minutes, assuming no difficulties arise.

1.2. Pupitre d'impression settings

The format key should be placed in the "Résultats" position. The lengths of the lines used in the various printouts are as follows :

semiprogram	: 32, 33, 39
table of designators	: 12
special printouts	: 18, 30

Since two of the semiprogram lines differ in length by only one character, it is necessary to short-circuit the end-of-line check contact (on the back of the typewriter) if that printout is desired.

1.3. Pupitre de commande settings

Register OB must be filled by hand, so as to put the input tape in position A and the output tape in position C.

The normal position for the Renvoi 1 and 2 switches is "Inhiber". The action controlled by these switches is as follows. RENVOI 1 : Normally, at the beginning of Part II, the routine records a blank word with serial 0002 on the output tape. At the end of Part II, the routine copies unchanged any additional words appearing on the input tape onto the output tape. (The copying ceases when a serial 0002 is read from the input tape) If Renvoi 1 is set to "Autor", both of these features are suppressed. The setting of Renvoi 1 may be changed while Part II is going on, so that only one of the features is suppressed. RENVOI 2 : Normally, during Part II, the semiprogram printout described in §1.2.2 is produced as the translation progresses. If Renvoi 2 is set to "Autor.", the semiprogram printout is suppressed, but at the end of the translation the table of designators and their corresponding addresses is printed. If both printouts are desired, set Renvoi 2 to "Inhiber"; then, at the end of the translation, start the computer at address 2555. If neither printout is desired, set Renvoi 2 to "Autor." and then stop the computer when the printing of the designator table begins.

"Org. Tambours" and "Org. Transfert" are automatically set to 6 and 9 respectively by the program, and similarly the decimal point is made "fixed".

1.4. Starting the routine

After OB, Renvoi 1, and Renvoi 2 are set as described in the previous section, the PCT routine can be started at order 0100. Note that after the completion of the "programme de transfert et de contrôle" or of the PCT routine itself, the computer is halted at address 0095; in this case it is only necessary to press "Programme Start".

In certain situations, it may be desirable to translate a sequence of pseudo-orders which are not preceded on the input tape by a "fin de ruban" si-

gnal. (They must end with a "fin de tranche signal", however.) In this case, the following word on the order drum :

0660 48000 0160 92000 0000

is replaced by :

0660 48000 0160 42000 0000.

Set the input tape to the proper starting point, set "Org. Tambours" to 6, and start the machine at order 0105. It will stop at the end of Part I, order 0660; now reset the input tape to its starting point, and then press "Programme Start" (without synchronizing).

2. Special Printouts

There are two types of special printouts which may occur during the translation to notify the operator of certain errors. The first type occurs only during Part I and is signalled by the printing of a series of 1's; the second occurs only during Part II and is signalled by a series of 2's.

2.1. Type 1 printouts

A type 1 printout indicates that the same designator has been assigned to more than one point in the pseudoprogram. The series of 1's which identifies the printout is followed by three 4-digit numbers, the first of which is the designator in question. The other two numbers are the two addresses in the translated program to which the designator corresponds; they are provided as a rough aid to locating the offending orders in the pseudoprogram. If the same designator has been assigned three or more times, a printout will occur for every extra assignment.

A type 1 printout causes the computer to stop : not immediately, but at order 0670 at the end of Part I. Thus, if any other designator has been assigned more than once, the fact will be signalled by an additional printout. Once the machine has stopped, the pseudoprogram tape must be corrected so that no designator is assigned twice. (If the corrections are not made, a "fin al." will occur in Part II.) After the pseudoprogram is corrected, the translation routine should be started again at address 0100.

2.2. Type 2 printouts

A type 2 printout indicates that a designator is used in the pseudo-program without ever having been defined by means of an "05" order. The series of 2's which identifies the printout is followed by one 4-digit number, which is the designator in question. Only one printout occurs for each undefined designator, no matter how many times it is used in the pseudoprogram.

Instead of stopping after a type 2 printout, the PCT routine simply uses the undefined designator as an address in the output program. For example, if the designator 1234 is never defined, the pseudo-order "40 1234" is translated as "40000 1234". Hence after the routine is finished, the programmer has the option of either correcting the wrong addresses in the output program, or inserting the missing designator into the pseudoprogram and then translating it again.

3. Stops

3.1. Stops without alarm

0095 : End of program

0670 : A type 1 printout has occurred during Part I. It is necessary to correct the pseudoprogram (so that each designator is assigned only once), and then to begin the translation again.

3.2. Stops with alarm ("final.")

0655 or 1445 : More than 860 designators have been used in the program, and there is not room for them all in the table. This is a most unlikely situation, and hence a machine failure is probably indicated.

1965 : The address in the output program corresponding to some designator, as determined in Part II, does not agree with the address which was listed in the table of designators during Part I. This would be caused by continuing into Part II after the occurrence of a type 1 printout in Part I; otherwise there has been a machine failure.

The "05" pseudo-order which has just been read is in register F, and its serial number on the input tape is in V. The absolute difference between the addresses determined in Parts I and II is contained in W. The actual addresses may be observed by operating the following orderword "pas à pas", after synchronizing and setting "Org. Tambour" to 6 :

1945 04166 0005 04076 0065

The first order adds into the groupe calculateur the address determined in Part I, and the second subtracts the one determined in Part II.

Assuming that the stop was caused by a machine failure, the first thing to try is to erase the output tape, start the program again at 0100, and see whether the failure recurs.

2030 : Some designator, read in Part II, cannot be found in the table of designators formed during Part I. This is obviously due to a machine failure. The "05" pseudo-order which has just been read is in F, and this serial number on the input tape is in V.

4. Modifications

In this section are discussed two simple modifications of the translation routine. Another possible modification, not affecting the main translation routine, would be the assignment of new locations to the ^{other} short routines (Chapter IV) or their elimination from the translation routine tape.

4.1. Prefixes 71-89

As noted in §3.4, prefixes 71-78 cause renvois to the elementary function subroutines while prefixes 79-89 cause renvois to location 8390, which contains a "fin al." instruction. This arrangement is easily modified. The translation routine stores the renvoi addresses used by 71-89 as the first four digits of number locations 0310, 0320, ..., 0490. Hence if it is desired to make prefix 79 cause a renvoi to order 6125, for example, it is only necessary to place 6125 in location 0390 on the number drum after the translation routine has been read in.

4.2. Storage requirements

The translation routine always uses number locations 0000-0605, and in addition locations 0610-9395 may be used for storing the table of designators. Since each table entry requires two numbers, about 880 designators may be used; this is far more than any normal program will use. However, to ensure that the permanent subroutine constants in locations 9400-9930 are not disturbed, the routine checks to make sure that the table is not extended as far as 9400. If it should be desired to put a lower limit on the table because of the storage of additional constants on the number drum, the following two orders should be altered :

0645	09096	0095	03076	9400
1440	03076	9400	07446	0000

In each of these orders, "9400" should be replaced by the address of the first number location which it is desired to leave untouched.

Chapter III : Details of the Translation Routine

1. Introduction

The Pseudocode Translation (PCT) Routine is made up of ten blocks, which are named α , β , 7, 5, 4, ξ , 1, 0, δ , and ρ . Block α forms Part I, and the remaining nine blocks make up Part II. In Part I, the input tape is read and the table of designators is formed. In Part II, each pseudo-order is read at the beginning of Block β . A few pseudo-orders are translated in β , but ordinarily for the translation there is a remvoi to one of Blocks 7, 5, 4, ξ , 1, 0, or δ . Finally there is a remvoi to Block ρ , where the recording and printing are done. There are certain exceptions to this general pattern; for example, orders 10 and 15 are translated partly in Block ξ and partly in Block 0.

Most of the routine is performed with fixed decimal point organization. However, during all printouts the machine has floating decimal point while the control words are transferred to register E.

During Part I the words in the (as yet non-existent) translated output are counted in order to compile the table of designators and their corresponding addresses. During Part II the count is of course repeated, but with the numbers stored in different registers, and the table of designators is checked. If any entry does not agree with the new count, a "final." occurs. This is the only check on the translation included in the PCT routine. It will detect many but not all types of machine error.

The ten blocks of the routine are described in §§4-13, and diagrams of each are given in §15.

2. Working Storage

Locations 0000-0095 on the order drum are used for working storage, and register Z_2 is similarly used. Each of the quantities stored in these locations is represented here by a symbol; these symbols are used throughout the remaining section of the chapter. A list of the locations, symbols, and quantities stored is given below.

0035	X_S	eighth digit of X , <u>in eighth position</u>
0040	T_1	first translated order for next order-word to be recorded
0045	T_2	second translated order for next order-word to be recorded
0050	U_1	"semi-order" corresponding to T_1
0055	U_2	"semi-order" corresponding to T_2
0060	L	serial number for next order-word to be recorded
0065	L_N	next serial number, for the following order word
0070	P	either = -1, or is the designator to be printed at right of next "semicode" printout
0075	P_N	either = -1, or is the designator to be printed with the following printout
0080	R	in Block ρ , stores control address for last quantity printed on each "semicode" line
0085	-	not used
0090	θ	= 0 normally; = -1 during translation of certain 70-89 orders
0095	Q	location of next entry to be added to table of designators.

The general symbols " T_i " and " U_i " are also often used in the following pages. The subscript $i=1$ when $l=0$, and $i=2$ when $l=5$.

3. "Programme de Transfert et de Contrôle", 0015-0090

This program is the same as that given in chapter I, §5. After transferring the second tranche of A onto the number drum, it checks the contents of the first two tranches (which may be in any order) against the order drum and number drum respectively.

Index register Z_1 holds the address of the latest word read from tape. Register Z_3 contains 0 while the orders are checked, and 1 while the numbers are checked. Register Z_4 normally contains 1, but at the end of each tranche it contains 0.

Both orders and numbers are checked by the same loop, involving orders 0030-0070. If a word fails to check, there is a stop at order 0095 and the conspicuous address "8888" may be seen in register N. The last word read from tape is in F, and its address is in Z_1 . If one wishes to check the remaining words, he need ordinarily only press "Programme Start".*

Coding					Notes
0002	00000	0000	00000	0000	End of tape
0015	90000	0006	54000	0000	Read in numbers; rewind tape; organize
0020	92000	0000	06800	0000	
0025	95000	0009	90000	0005	
0030	52000	0000	06900	0001	Read, $A \rightarrow F$, $1 \rightarrow Z_4$. If f.t., $0 \rightarrow Z_4$
0035	48000	0040	06900	0000	
0040	02566	0000	06646	0000	If words are not equal, Rv 0090
0045	04168	0000	02078	0000	
0050	04169	0000	02079	0000	
0055	07146	0000	00082	0000	
0060	02166	0000	07446	0000	
0065	41000	0090	03476	0000	
0070	07446	0000	41000	0030	If not f.t., Rv 0030
0075	03376	0000	07446	0000	If program finished, Rv 0095
0080	41000	0095	06800	0001	
0085	90000	0006	40000	0030	To check numbers, Rv 0030
0090	42000	8888	40000	0030	FIN

4. Block α , 0095-0675

Block α makes up Part I of the PCT Routine. Its function is to count the order-words which will appear in the translated program, and to make up a "table of designators" containing each designator with its corresponding order address. If any designator is defined more than once, quantity O^1 is changed from -1 to 0; this causes the machine to stop at the end of the block (order 0670).

* If the word which failed to check was the last word of the first tranche, one must change Z_3 from 0 to 1 and "Org. Tambours" from 5 to 6 before starting the machine.

Block α includes the Table Look-Up or TLU subroutine (orders 0530-0570), which is used for finding a given entry in the table of designators. This subroutine is used frequently during Part II.

Coding					Notes
0095	42000	0000	00000	0000	Program stop
0100	92000	0000	90000	0006	Rewind tape A. Organize machine
0105	95000	0009	00021	0000	
0110	07346	0000	03076	0001	$-1 \rightarrow 0'$, $0 \rightarrow 1'$, $15 \rightarrow L'$, $0610 \rightarrow Q$
0115	09046	0090	09046	0005	
0120	06700	0015	03066	0610	
0125	09046	0095	00000	0000	
0130	00000	0000	00000	0000	
0135	52000	0000	02066	0000	READ A, $\rightarrow F$. If $X_P \neq 38$, Rv 0165
0140	00043	0000	03036	1100	
0145	03076	3800	07246	0000	
0150	02276	0000	07446	0000	
0155	41000	0165	06700	0095	$0095 \rightarrow L'$
0160	52000	0000	00000	0000	READ A, $\rightarrow F$
0165	02066	0000	09096	0000	Store X , X_S , X_A , X_P
0170	00033	0000	04036	0220	
0175	09046	0025	02066	0000	
0180	00033	0000	03036	0100	
0185	07246	0000	02266	0000	
0190	09046	0015	02066	0000	
0195	00043	0000	03036	1100	
0200	09096	0010	03076	3900	If $X_P = 39$, Rv 0470
0205	07246	0000	02266	0000	
0210	03076	0001	07446	0000	
0215	41000	0470	00000	0000	
0220	00000	0000	04066	0010	If $X_P = 05$, Rv 0490
0225	03076	0500	07246	0000	
0230	02266	0000	03076	0001	
0235	07446	0000	41000	0490	
0240	04066	0010	03076	4500	If $X_P = 45$, Rv 0380
0245	07246	0000	02266	0000	
0250	03076	0001	07446	0000	
0255	41000	0380	00000	0000	
0260	00000	0000	00000	0000	
0265	00000	0000	00000	0000	
0270	00000	0000	00000	0000	

0275	04076	0010	03066	9000	If $89 \Rightarrow X_P \geq 70$, Rv 0340
0280	07446	0000	41000	0300	
0285	04076	0010	03066	7000	
0290	07446	0000	41000	0340	
0295	00000	0000	00000	0000	
0300	04066	0025	03076	2500	If $1 \leq X_S \leq 4$, Rv 0350
0305	07246	0000	02266	0000	
0310	03076	2000	07446	0000	
0315	41000	0350	04076	0005	If $1' \neq 0$, Rv 0350
0320	07446	0000	41000	0350	
0325	03066	0005	09046	0005	$5 \rightarrow 1'$, Rv 0660
0330	40000	0660	00000	0000	
0340	04076	0005	07446	0000	If $1' \neq 0$, Rv 0365
0345	41000	0365	00000	0000	
0350	01200	0005	09046	0005	$L' + 5 \rightarrow L'$, $0 \rightarrow 1'$. Rv 0660
0355	40000	0660	00000	0000	
0365	01200	0010	09046	0005	$L' + 10 \rightarrow L'$, $0 \rightarrow 1'$. Rv 0660
0370	40000	0660	00000	0000	
0380	04066	0005	06296	0000	$(X_P = 45.)$ Count one pseudo-order; Rv 0300
0385	03066	0005	00043	0000	
0390	03036	0001	09046	0005	
0395	40000	0300	00000	0000	
0470	00000	0000	04066	0015	$(X_P = 39.)$ $X_A \rightarrow L$, $0 \rightarrow 1$. Rv 0660
0475	06746	0000	09046	0005	
0480	40000	0660	00000	0000	
0490	04066	0005	06246	0000	$(X_P = 05.)$ $L' + 1' \rightarrow L'$, $0 \rightarrow 1'$.
0495	09046	0005	00000	0000	
0500	04066	0015	03076	0001	If $X_A = 0$, Rv 0660
0505	07446	0000	41000	0660	
0510	00000	0000	00000	0000	
0515	06800	0580	06900	0630	$0580 \rightarrow Z_3$, $0630 \rightarrow Z_4$
0520	00000	0000	00000	0000	
0525	00000	0000	00000	0000	
0530	06600	0600	00000	0000	$0600 \rightarrow Z_1$
0535	04166	0000	04076	0015	If entry = X_A , Rv Z_3
0540	07246	0000	02266	0000	
0545	03076	0001	07446	0000	
0550	41300	0000	01100	0010	$Z_1 + 10 \rightarrow Z_1$. If $Z_1 \neq 0$, Rv 0535
0555	03166	0000	04076	0095	
0560	07246	0000	02276	0000	
0565	07446	0000	41000	0535	$(X_A \text{ not in table})$ Rv Z_4
0570	40400	0000	00000	0000	

TLU
Sub-
routine

0580	00011	0000	09046	0090	(Designator defined twice.) $0 \rightarrow \theta'$. Type 1 printout. Rv 0660
0585	04066	0140	07146	0000	
0590	04066	0180	04066	0170	
0595	05046	0000	04076	0110	
0600	07146	0000	04066	0015	
0605	05046	0000	00000	0000	
0610	04166	0005	05046	0000	
0615	03266	0000	05046	4000	
0620	00021	0000	40000	0660	
0630	04066	0015	09146	0000	Add to table $X_A: L^r$. $Q + 10 \rightarrow Q$.
0635	03266	0000	09146	0005	
0640	04066	0095	03066	0010	
0645	09096	0095	03076	9400	If $Q = 9400$, FIN AL.
0650	07446	0000	41000	0660	
0655	47000	0000	00000	0000	
0660	48000	0160			If not f.t., Rv 0160
			92000	0000	Rewind tape A
0665	04066	0090	07446	0000	If $\theta' = 0$, FIN
0670	41500	0000	42000	0000	
0675	44000	0705	40000	0700	If Renvoi 1 is set, Rv 0705. Otherwise Rv 0700

5. Block β , 0700-1115

Block β is the first block of Part II. Its main function is to cause a renvoi to one of blocks 7, 5, 4, ϵ , 1, 0, or δ , where the current pseudo-order will be translated. However, "Alteration" orders, as well as those with prefixes 36-38, are translated in block β .

At order 0960, when $X_S = 8$, there is a renvoi to 1090 and thence to the TLU subroutine. If the desired table entry is found, the machine returns to order 1100 and shortly afterwards to 0965. If the entry is not found, the machine goes to order 1395 in Block 4; there is a type 2 typeout, etc., followed by a renvoi to 1100.

Coding					Notes
0700	07146	0000	08046	0002	Serial 0002 $\rightarrow$ tape C
0705	03066	0015	09046	0060	0015 $\rightarrow$ L. 0 $\rightarrow$ 1, T ₁ , T ₂ , U ₁ , U ₂ , 0, -1 $\rightarrow$ P, P _N
0710	06700	0000	09046	0040	
0715	09046	0045	09046	0050	
0720	09046	0055	09046	0090	
0725	03076	0001	09096	0070	
0730	09046	0075	00000	0000	
0735	00000	0000	00000	0000	
0740	52000	0000	02066	0000	READ A, $\rightarrow$ F. If X _P $\neq$ 38, Rv 0805
0745	00043	0000	03036	1100	
0750	03076	3800	07246	0000	
0755	02276	0000	07446	0000	
0760	41000	0805	00011	0000	Transfer words 0015-0090 from order drum to C
0765	90000	0005	06600	0015	
0770	04168	0000	00049	0000	
0775	04169	0000	08149	0000	
0780	01100	0005	03176	9905	
0785	07446	0000	41000	0770	
0790	03166	0000	09046	0060	
0795	90000	0006	00021	0000	
0800	52000	0000	00000	0000	READ A, $\rightarrow$ F.
0805	02066	0000	09096	0000	Store X, X _P , X _{P2} , X _A , X _I , X _S .
0810	00043	0000	03036	1100	
0815	09096	0010	00033	0000	
0820	03036	1000	09046	0005	
0825	02066	0000	00033	0000	
0830	03036	0100	00043	0000	
0835	03036	1111	09046	0015	
0840	02066	0000	00043	0000	
0845	04036	0220	09046	0020	
0850	02066	0000	00043	0000	
0855	04036	0230	09096	0035	
0860	00033	0000	04036	0220	
0865	09046	0025	00000	0000	
0870	04066	0010	03076	3900	If X _P = 39, Rv 1900 (Block 8)
0875	07246	0000	02266	0000	
0880	03076	0001	07446	0000	If X _P = 05, Rv 1915 (Block 8)
0885	41000	1900	04066	0010	
0890	03076	0500	07246	0000	
0895	02266	0000	03076	0001	
0900	07446	0000	41000	1915	

0905	02066	0000	00000	0000	$X \rightarrow U_1$
0910	09246	0050	04066	0010	If $89 \geq X_P \geq 70$, Rv 1130B (Block 7)
0915	03076	7950	07246	0000	
0920	02266	0000	03076	1000	
0925	07446	0000	41000	1130	
0930	04066	0010	03076	4450	If $49 \geq X_P \geq 40$, Rv 1280 (Block 4)
0935	07246	0000	02266	0000	
0940	03076	0500	07446	0000	
0945	41000	1280	04066	0025	
0950	03076	8000	07246	0000	If $X_S = 8$, Rv 1090
0955	02266	0000	03076	0001	
0960	07446	0000	41000	1090	
0965	04076	0010	03066	9000	
0970	07446	0000	41000	1500	If $X_P \geq 90$, Rv 1500 (Block 4)
0975	04076	0010	03066	5000	
0980	07446	0000	41000	1210	
0985	00000	0000	00000	0000	
0990	04066	0005	03076	5000	If $X_{P2} \geq 5$, subtract 5; result $\rightarrow F$. If $F = 0$, Rv 1520 (Block 8)
0995	07496	0000	41000	1005	
1000	03076	5000	00000	0000	
1005	03066	5000	07096	0000	
1010	07446	0000	41000	1520	If $X_P \leq 09$, Rv 1730 (Block 0)
1015	04066	0010	03076	1000	
1020	07446	0000	41000	1730	
1025	04066	0010	03076	2000	
1030	07446	0000	41000	1660	If $X_P \leq 19$, Rv 1660 (Block 1)
1035	04076	0010	03066	3600	
1040	07446	0000	41000	1065	
1045	04066	0010	03036	0001	
1050	04066	0015	04066	0020	(X _P is "Alteration".) Shift X _P right 4 places; + X _A + X _I $\rightarrow$ T ₁ . Rv 2070 (Block p)
1055	09246	0040	40000	2070	
1065	02066	0000	04036	0200	
1070	04066		04066	0015	
1075	04066	0020	09246	0040	(X _P is 36-38.) Shift F right 5; +.00004 + X _A + X _I $\rightarrow$ T ₁ . Rv 2070 (Block p)
1080	40000	2070	00000	0000	
1090	06800	1100	06900	1395	
1095	40000	0530	00000	0000	
1100	04166	0005	09096	0015	Replace X _A with table entry. Change U ₁ . Rv 0965
1105	03036	0100	04066	0010	
1110	04066	0020	09246	0050	
1115	40000	0965	00000	0000	

6. Block 7, 1130-1205

Block 7 translates pseudo-orders 70-89. Each of these must correspond to one complete order-word in the translated program, so orders 1130-1135 check whether the previous translated order-word has been completed. If not, there is a renvoi (orders 1195-1205) to record the word (thus effectively inserting a blank order). Then, by means of orders 2470-2475 in Block P, the machine returns to 1130 without first reading from A.

At orders 1140-1145, the order " $M \rightarrow Z_4$ " is placed in T_1 and U_1 . If $X_P = 70$, the translation is completed in Block 4. Otherwise an order for a renvoi to the appropriate elementary function subroutine is placed in T_2 and U_2 .

Coding					Notes
1130	09046	0090	03276	0000	$0 \rightarrow 0$. If $1 \neq 0$, Rv 1195
1135	07446	0000	41000	1195	
1140	04066	0245	09046	0040	Fill T_1 and U_1 . $5 \rightarrow 1$. Shift $X_P = 70$ right 1; $\rightarrow Z_1$. If $X_P = 70$, Rv 1365 (Block 4), having set $X_P = 40$.
1145	04066	0300	09046	0050	
1150	06700	0005	03076	7000	
1155	04066	0010	03036	1000	
1160	06696	0000	07446	0000	
1165	03066	4000	09046	0010	Use Z_1 to look up address of elementary function subroutine in 0310, 0320, ..., 0490. Add "Renvoi" order, and fill T_2 and U_2 . Rv 2070 (Block P)
1170	41000	1365	04166	0300	
1175	04066	0285	09096	0045	
1180	03036	0100	03066	4000	
1185	09046	0055	40000	2070	$0 \rightarrow T_2$, $-1 \rightarrow 0$, $0 \rightarrow Z_4$. Rv 2160 (Block P)
1195	09246	0050	03076	0001	
1200	09046	0090	06900	0000	
1205	40000	2160	00000	0000	

7. Block 5, 1210-1270

Block 5 translates pseudo-orders 50-69. ($X_P - 50$) is placed in an index register and thus used to read the translated prefix from one of locations 0305, 0315, ..., 0495 on the number drum. Certain of the prefixes should not be augmented by an index digit; these prefixes are stored with a minus sign and the routine treats them accordingly.

The only unusual feature of Block 5 is that since the index digit accompanying pseudo-orders 50-69 should always be less than 5, a digit which (because of a coding error) is ≥ 5 is automatically decreased by 5.

Coding					Notes
1210	04066	0010	03076	5000	Use Xp-50 to look up translated prefix in 0305, 0315, ..., 0495. If the prefix has a minus sign, Rv 1260.
1215	03036	1000	06646	0000	
1220	04166	0305	07446	0000	
1225	41000	1260	04066	0020	If $X_I \geq 5$, subtract 5; leave in g.c.
1230	04076	0280	07446	0000	
1235	41000	1255	00000	0000	
1240	04266	0050	04076	0280	
1245	09246	0050	04076	0280	
1250	00000	0000	00000	0000	
1255	04066	0020	00000	0000	
1260	04066	0015	00032	0000	+ X_A + translated prefix $\rightarrow T_1$. Rv 2070 (Block ρ)
1265	04166	0305	09246	0040	
1270	40000	2070	00000	0000	

8. Block 4, 1280-1515

Block 4 translates pseudo-orders 40-49 and 90-99. In the case of the latter group the block is entered at order 1500. The block also completes the translation of pseudo-order 70, in which case the entrance is at 1365.

In the case of the pseudo-order 45, a "ch" order must be inserted first of all. If this completes a translated order-word, at 1310 there is a renvoi to Block ρ in order to record the word. Then, by means of orders 2480-2485 in that block, the machine returns to order 1325.

Orders 1395-1445, which cause a type 2 printout, are performed when the TLU subroutine fails to find a designator in the table. These orders may be used when the machine has gone to the TLU subroutine from Block β , as well as from the present block, and therefore the renvoi in 1445 may be either to 1100 (Block β) or to 1450.

Coding					Notes
1280	04066	0010	03076	4500	If $X_P \neq 45$, Rv 1335
1285	07246	0000	02276	0000	
1290	07446	0000	41000	1335	
1295	04066	0295	09246	0040	Put "ch" order into T_1 and U_1 . If $1 \neq 0$, Rv 2160 (Block ρ), with $X_P = -1$.
1300	04066	0500	09246	0050	
1305	03276	0000	07496	0000	
1310	09046	0010	4100	2160	
1315	06700	0005	00000	0000	$5 \rightarrow 1$
1320	00000	0000	00000	0000	
1325	03066	4100	09046	0010	$41 \rightarrow X_P, 1 \rightarrow X_{P2}$
1330	03066	1000	09046	0005	
1335	04066	0005	03076	5000	If $X_P = 42$ or 47 , Rv 1500
1340	07496	0000	03066	3000	
1345	41000	1350	03076	5000	
1350	07246	0000	02266	0000	
1355	03076	0001	07446	0000	
1360	41000	1500	00000	0000	
1365	04066	0025	03076	8000	
1370	07246	0000	02266	0000	If $X_S = 8$, Rv 1450, with $Z_1 = 10$
1375	03076	0001	07446	0000	
1380	06600	0010	4100	1450	
1385	06800	1450	06900	1396	
1390	40000	0530	00000	0000	$1450 \rightarrow Z_3, 1395 \rightarrow Z_4$. Rv TLU
1395	00011	0000	04066	0100	(Couldn't find entry.) Type 2 printout. Add new entry to table. If $Q = 9400$, PIN AL; otherwise Rv Z_3 , which = 1450 or 1100
1400	07146	0000	05046	5000	
1405	04066	0140	07146	0000	
1410	04066	0185	04066	0170	
1415	05046	0000	04076	0110	
1420	07146	0000	00021	0000	
1425	04066	0015	05096	4000	
1430	09196	0000	09146	0005	
1435	03166	0010	09096	0095	
1440	03076	9400	07446	0000	
1445	41300	0000	47000	0000	
1450	04066	0020	04076	0280	
1455	07146	0000	00032	0000	
1460	02176	0000	07446	0000	
1465	04166	0005	41500	0015	
1470	06946	0000	00000	0000	From g.c. subtract $L+5$, modulo 10000
1475	04076	0060	03066	9995	
1480	06446	0000	03466	0000	

1485	09096	0015	03036	0100	G.c. $\rightarrow X_A$. Shift new X_A right 2; + X_P + $X_I \rightarrow U_I$
1490	04066	0010	04066	0020	
1495	09246	0050	00000	0000	
1500	04066	0010	03036	1000	Shift X_{P1} to 9th digit position, and X_{P2} to 8th; + $X_I + X_A \rightarrow T_1$. Rv 2070 (Block p)
1505	04066	0005	04036	0220	
1510	04066	0020	04066	0015	
1515	09246	0040	40000	2070	

9. Block ξ , 1520-1650

Block ξ translates pseudo-orders 00, 10, 15, 20, 25, 30, and 35. In the case of 10 and 15 there is a rerout to Block 0 where the translation is completed.

Coding					Notes
1520	04066	0010	00000	0000	If $X_P = 00$, Rv 2070 (Block p), after $X_A \rightarrow T_1$
1525	00000	0000	03076	0001	
1530	07446	0000	04066	0015	
1535	09246	0040	4100	2070	If $X_S \neq 6$ or 7, Rv 1570
1540	09246	0040	04066	0025	
1545	03076	6500	07246	0000	
1550	02276	0000	03066	1000	
1555	07446	0000	41000	1570	
1560	04066	0205	09246	0040	.00003 $\rightarrow T_1$
1565	00000	0000	00000	0000	
1570	04066	0005	04036	0200	Shift X_{P2} left 5; + .000064 + T_1 $\rightarrow T_1$
1575	04066	0225	04266	0040	
1580	09246	0040	04066	0025	If $F_S = 6$, Rv 1645
1585	03076	6000	07246	0000	
1590	02266	0000	03076	0001	
1595	07446	0000	41000	1645	
1600	00000	0000	00000	0000	
1605	04066	0290	04266	0040	.0000 0005 + $T_1 \rightarrow T_1$
1610	09246	0040	04076	0010	
1615	03066	3000	07446	0000	If $X_P \geq 30$, Rv 1635
1620	41000	1635	04076	0010	
1625	03066	2000	07446	0000	If $X_P \geq 20$, Rv 1640. Otherwise Rv 1805 (Block 0)
1630	41000	1640	40000	1805	
1635	04066	0255	00000	0000	+ .0000 0003
1640	04066	0020	00000	0000	
1645	04066	0015	04266	0040	+ X_I + $X_A + T_1 \rightarrow T_1$. Rv 2070 (Block p)
1650	09246	0040	40000	2070	

10. Block 1, 1660-1680

Block 1 translates orders 11-14 and 16-19.

Coding					Notes
1660	04066	0005	03036	0100	Use X_{P2} to look up g.c. part of prefix in 0515, 0525, ..., 0595; + .0000 0003 + X_I + $X_A \rightarrow T_1$, Rv 2070 (Block p)
1665	06646	0000	04166	0505	
1670	04066	0255	04066	0020	
1675	04066	0015	09246	0040	
1680	40000	2070	00000	0000	

11. Block 0, 1730-1865

Block 0 translates pseudo-orders 01-04 and 06-09. It also completes the translation of 10 and 15, in which case it is entered at order 1805.

Coding					Notes
1730	02076	0000	03066	3000	If F (which = X_{P2} reduced modulo 5) = 3 or 4, Rv 1790
1735	07446	0000	41000	1790	
1740	04066	0025	03076	6500	If $X_S \neq 6$ or 7, Rv 1790
1745	07246	0000	02276	0000	
1750	03066	1000	07446	0000	Shift F right 1, +(X_S-5); shift sum right 4, +.0000 75 $\rightarrow T_1$ Rv 1805
1755	41000	1790	02066	0000	
1760	03036	1000	04066	0025	
1765	03076	5000	03036	0001	
1770	04066	0235	09246	0040	
1775	00000	0000	00000	0000	
1780	40000	1805	00000	0000	
1790	04066	0005	03036	0100	Use X_{P2} to look up g.c. part of prefix in 0515, 0525, ..., 0595; $\rightarrow T_1$
1795	06646	0000	04166	0505	
1800	09246	0040	00000	0000	
1805	04066	0015	00033	0000	Shift X_A left 3 to isolate last digit; $\rightarrow F$. If digit = 0 or 5, Rv 1855
1810	03036	0010	07096	0000	
1815	03076	5000	07496	0000	
1820	41000	1825	03076	5000	
1825	03066	5000	07446	0000	
1830	41000	1855	02066	0000	Use F to look up first part of prefix in 0510, 0520, ..., 0590; + $T_1 \rightarrow T_1$. Rv 2070 (Block p)
1835	03036	0100	06646	0000	
1840	04166	0500	04266	0040	
1845	09246	0040	40000	2070	
1855	04066	0265	04066	0020	.0000 0004 + X_I + X_A + $T_1 \rightarrow T_1$. Rv 2070 (Block p)
1860	04066	0015	04266	0040	
1865	09246	0040	40000	2070	

12. Block δ , 1900-2030

Block δ deals with pseudo-orders 05 and 39. For 39, it is only necessary to replace L_N by X_A . For 05, the block checks that the table of designators formed in Part I is correct. If the address listed in the table for the designator X_A does not check with L_N , the "fin al." of order 1965 occurs. If X_A cannot be found in the table at all, the "fin al." of 2030 occurs.

If the previous translated order-word has not been recorded, there occurs a renvoi at order 1975 in order to record it. This effectively inserts one blank order in the word.

In certain situations it can be necessary to record a "fin de tranche" or "fin de groupe" indication when the previous translated order-word has already been recorded. In this case, orders 2015-2025 provide for a blank word to be recorded with the serial number (0000 + i), where i is the appropriate indication. The normal sequence of order-words is then resumed.

Coding					Notes
1900	04066	0015	09046	0065	(X _P = 39.) $X_A \rightarrow L_N$. Rv 1970
1905	40000	1970	00000	0000	
1915	04066	0060	03266	0000	(X _P = 05.) $L + 1 \rightarrow L_N$. If $X_A = 0$, Rv 1970
1920	09046	0065	04066	0015	
1925	03076	0001	07446	0000	
1930	41000	1970	04066	0015	$X_A \rightarrow P_N$. 1945 $\rightarrow Z_3$, 2030 $\rightarrow Z_4$. Rv TLU.
1935	09046	0075	06800	1945	
1940	06900	2030	40000	0530	
1945	04166	0005	04076	0065	If table entry $\neq L_N$, FIN AL.
1950	07246	0000	02266	0000	
1955	03076	0001	07446	0000	
1960	41000	1970	00000	0000	
1965	47000	0000	00000	0000	
1970	03276	0000	07446	0000	
1975	41000	2080	04066	0025	If $1 \neq 0$, Rv 2080 (Block P) If $X_S = 1, 2, 3$, or 4, Rv 2015
1980	03076	2500	07246	0000	
1985	02266	0000	03076	2000	
1990	07446	0000	41000	2015	

1995	04066	0075	09046	0070	$P_N \rightarrow P, -1 \rightarrow P_N, L \rightarrow L. \text{ If not f.t.,}$ $Rv\ 0600\ (\text{Block } \beta)$
2000	03076	0001	09046	0075	
2005	04066	0065	09046	0060	
2010	48000	0800	00000	0000	
2015	03076	0001	09046	0070	$-1 \rightarrow P, 0 \rightarrow L, 5 \rightarrow L. Rv\ 2030\ (\text{Block } \rho)$
2020	09046	0060	06700	0005	
2025	40000	2080	00000	0000	
2030	47000	0000	00000	0000	FIN AL.

13. Block ρ . 2070-2600

Block ρ is the final block of Part II. Renvois from block enter at order 2080; and special renvois from Blocks 7 and 4 enter at 2160. Ordinarily, however, Block ρ is entered at order 2070 after the translation of each pseudo order. The address counters L, l are advanced, and if no recording is to be made there is a renvoi at order 2135 to Block β , where the next pseudo-order is read from tape. However, if an order-word has been completed, or if a "fin de tranche" or "fin de groupe" indication must be made, the the order-word is recorded at order 2190.

Orders 2200-2440 provide for the "semicode" printout. This sequence is quite complicated, since a variety of control numbers must be used in order to print only non-zero index digits, insert special digits for orders having $X_8 = 6$ or 7, and preserve correct spacing. The address of the order-word is printed at 2240. The subsequence 2255-2365 is performed twice in order to print the two orders. Index register Z_3 is used to select the control number for the order about to be printed, while Z_4 accumulates a quantity for the selection of the next control number. The register "P" (0070) is negative unless it contains a designator to be printed; this printing is done by orders 2420-2440.

Orders 2445-2465 reset various registers in preparation for the next order-word. Orders 2470-2485 cause renvois to Blocks 7 or 4 if the present block was entered at 2160, and otherwise cause a return to Block β .

The remaining orders are performed only after the whole pseudoprogram has been translated. Orders 2505-2545 copy the remainder of tape A onto tape C, and 2555-2575 print out the table of designators.

Coding					Notes
2070	00000	0000	04066	0060	$L + 1 \rightarrow L_N$
2075	03266	0000	09046	0065	
2080	04066	0025	03076	2500	If $X_S = 1, 2, 3$, or 4, Rv 2140
2085	07246	0000	02266	0000	
2090	03076	2000	07446	0000	
2095	41000	2140	00000	0000	If not f.t., Rv 2125
2100	48000	2125	06900	0003	$3 \rightarrow Z_4$. If $X_S \neq 5$, Rv 2150
2105	04066	0025	03076	5000	
2110	07246	0000	02276	0000	
2115	07446	0000	41000	2150	
2120	06900	0000	40000	2150	$0 \rightarrow Z_4$ Rv 2150
2125	06900	0000	03276	0000	$0 \rightarrow Z_4$ If $1 = 5$, Rv 2170
2130	07446	0000	41000	2170	
2135	06700	0005	40000	0800	$5 \rightarrow 1$, Rv 0800 (Block β)
2140	04066	0025	03036	0010	X_S , shifted right 3, $\rightarrow Z_4$
2145	06946	0000	00000	0000	
2150	03276	0000	07446	0000	If $1 = 5$, Rv 2170
2155	41000	2170	00000	0000	
2160	00000	0000	04066	0060	$L + 5 \rightarrow L_N$
2165	03066	0005	09046	0065	
2170	06700	0000	00000	0000	$0 \rightarrow 1$. $Z_4 + L \rightarrow Z_4$. RECORD T_1 and $T_2 \rightarrow C$, with serial Z_4
2175	04066	0060	06446	0000	
2180	07146	0000	04066	0040	
2185	00049	0000	04066	0045	
2190	08449	0000	49000	2445	If Renvoi 2 is set, omit semicode printout and Rv 2445
2195	00000	0000	00000	0000	
2200	00000	0000	04066	0060	If $L + 5$ ends in 50, Rv 2225
2205	03066	0005	03036	2000	
2210	00033	0000	03036	0010	
2215	03076	0001	07446	0000	
2220	41000	2225	03066	1000	+ 1000
2225	03066	4000	09046	0080	+ 4000 $\rightarrow R$. PRINT Z_4 . $0 \rightarrow Z_1, Z_3$
2230	00011	0000	04066	0105	
2235	07146	0000	03466	0000	
2240	04066	0170	05046	0000	
2245	06600	0000	06800	0000	
2250	00021	0000	00000	0000	
2255	06900	0000	04266	0050	$0 \rightarrow Z_4$. If seventh digit of $U_1 = 0$ Rv 2280
2260	00043	0000	04036	0220	
2265	04076	0230	07446	0000	
2270	41500	0005	01300	0010	$Z_3 + 10 \rightarrow Z_3$, $5 \rightarrow Z_4$
2275	06900	0005	00000	0000	

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2280	04266	0050	00033	0000	If eighth digit of $U_1 = 0$, Rv 2350
2285	04036	0220	03036	0100	
2290	07096	0000	00000	0000	
2295	03076	0065	07246	0000	
2300	02276	0000	03066	0010	
2305	07446	0000	41000	2350	
2310	04266	0050	00033	0000	Rearrange U_1 for printout, $Z_3 + 20 \rightarrow Z_3$
2315	03036	0100	03036	0010	
2320	07146	0000	04266	0050	
2325	00043	0000	03036	1100	
2330	02166	0000	02066	0000	
2335	03076	0050	09246	0050	
2340	01300	0020	00000	0000	
2345	00000	0000	00000	0000	
2350	00011	0000	04366	0120	Select control number with $Z_3 \rightarrow E$. PRINT U_1
2355	07146	0000	04266	0050	
2360	04066	0170	05146	0000	
2365	00021	0000	03276	0000	
2370	07446	0000	41000	2410	If $1 = 5$, Rv 2410 $5 \rightarrow 1$. $Z_4 \rightarrow Z_3$. If P is positive, Rv 2255
2375	06700	0005	03466	0000	
2380	06846	0000	04076	0070	
2385	03076	0001	07446	0000	
2390	41000	2255	04066	0080	$R \rightarrow Z_1$. Rv 2255
2395	06646	0000	40000	2255	
2410	06700	0000	04066	0070	$0 \rightarrow 1$. If P is negative, Rv 2445
2415	07446	0000	41000	2445	
2420	00011	0000	04476	0110	Select control number $Z_4 \rightarrow E$. $R \rightarrow Z_1$. PRINT P
2425	07146	0000	04066	0080	
2430	06646	0000	04066	0070	
2435	05146	0000	00021	0000	
2440	00000	0000	00000	0000	
2445	09046	0040	09046	0045	$0 \rightarrow T_1, T_2, U_1, U_2$. $L_N \rightarrow L$, $P_N \rightarrow P$, $-1 \rightarrow P_N$
2450	09046	0050	09046	0055	
2455	04066	0065	09046	0060	
2460	04066	0075	09046	0070	
2465	03076	0001	09046	0075	
2470	04066	0090	07446	0000	If θ is negative, Rv 1130 (Block 7)
2475	41000	1130	00000	0000	
2480	04066	0010	07446	0000	If X_p is negative, Rv 1325 (Block 4)
2485	41000	1325	48000	0800	
2490	00000	0000	00000	0000	If not f.t., Rv 0800 (Block 3)
2495	00011	0000	44000	2550	
2500	00000	0000	00000	0000	Floating point org. If Renvol 1 is set, Rv 2550

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2505	95000	0004	52000	0000	READ A, $\rightarrow$ F. If serial = 0002, Rv 2550
2510	02566	0000	06696	0000	
2515	03076	0002	07246	0000	
2520	02266	0000	03076	0001	
2525	07446	0000	41000	2550	
2530	02068	0000	00049	0000	RECORD F $\rightarrow$ C, TRANSFER rest of tran- che, A $\rightarrow$ C. Rv 2505
2535	02069	0000	08149	0000	
2540	95000	0000	53000	0000	
2545	40000	2505	00000	0000	
2550	49500	0000	40000	0095	If Renvoi 2 is not set, Rv 0095 (FIN).
2555	04066	0100	07146	0000	One carriage return. 0610 $\rightarrow$ Z ₁
2560	05046	5000	06600	0610	
2565	03176	0000	04066	0095	If Z ₁ = Q. Rv 0095 (FIN)
2570	07446	0000	41000	0095	
2575	04076	0105	07146	0000	PRINT one line of designator table. Z ₁ + 10 $\rightarrow$ Z ₁ . Rv 2565
2580	04166	0000	04066	0170	
2585	05046	0000	04166	0005	
2590	04066	0170	05046	5000	
2595	01100	0010	40000	2565	
2600	02016	0000	08146	0000	

The remaining orders in the first tranche of the PCT routine tape, with serial numbers 7940-8168, are given in Chapter 4.

14. Constants

The following constants make up the second and last tranche of the PCT routine tape.

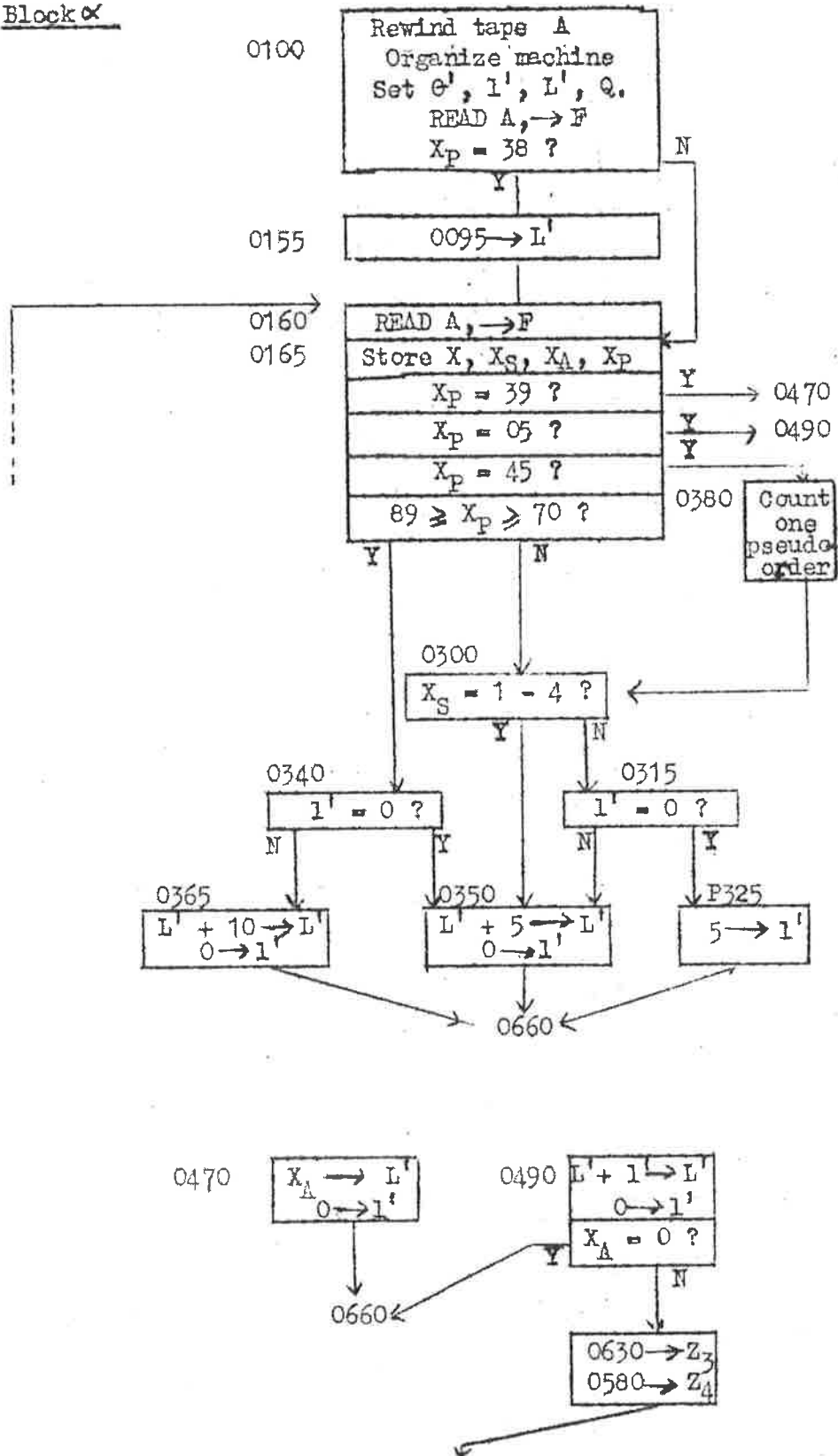
0100	- .44444	44444	44444	+ 44
0105	+ .40000	44444	44444	+ 44
0110	+ .00004	44444	44444	+ 44
0115	+ .10000	44444	44444	+ 44
0120	+ .400	1000	44444444	+ 44
0125	+ .410	1000	44444444	+ 44
0130	- .400	1000	14444444	+ 44
0135	- .410	1000	14444444	+ 44
0140	+ .4000	0000	44444444	+ 44
0145	+ .4100	0000	44444444	+ 44
0150	- .4000	0000	14444444	+ 44
0155	- .4100	0000	14444444	+ 44
0160	0			
0165	0			
0170	+ .0			+ .01
0175	0			

0180	+.11111	11111
0185	+.22222	22222
0190	+.33333	33333
0195	+.55555	55555
0200	+.0000	1
0205	+.0000	3
0210	+.0000	01
0215	+.0000	4
0220	+.0000	001
0225	+.0000	64
0230	+.0000	0001
0235	+.0000	75
0240		0
0245	+.0000	0091
0250		0
0255	+.0000	0003
0260		0
0265	+.0000	0004
0270		0
0275	+.0000	0009
0280	+.0000	005
0285	+.0000	00004
0290	+.0000	0005
0295	+.0000	64470
0300	+.68	0000 4
0305	+.0000	00005
0310	+.8265	
0315	+.0000	00125
0320	+.8395	
0325	+.0000	00025
0330	+.8575	
0335	+.0000	00035
0340	+.8790	
0345	+.0000	00045
0350	+.9060	
0355	+.0000	00055
0360	+.9105	
0365	+.0000	00175
0370	+.9530	
0375	+.0000	00075
0380	+.9575	
0385	+.0000	00085
0390	+.8390	
0395	+.0000	00095

0400	+.8390	
0405	+.0000	11000
0410	+.8390	
0415	+.0000	00560
0420	+.8390	
0425	+.0000	64560
0430	+.8390	
0435	+.0000	00010
0440	+.8390	
0445	+.0000	64060
0450	+.8390	
0455	+.0000	12000
0460	+.8390	
0465	-.0000	00770
0470	+.8390	
0475	+.0000	69560
0480	+.8390	
0485	+.0000	00510
0490	+.8390	
0495	+.0000	69060
0500	+.10	0004
0505	0	
0510	+.0000	00120
0515	+.0000	61
0520	+.0000	00020
0525	+.0000	62
0530	+.0000	00320
0535	+.0000	63
0540	+.0000	00420
0545	+.0000	53
0550	0	
0555	0	
0560	+.0000	00020
0565	+.0000	66
0570	+.0000	00220
0575	+.0000	67
0580	+.0000	00820
0585	+.0000	68
0590	+.0000	00920
0595	+.0000	58
0600	0	
0608	0	

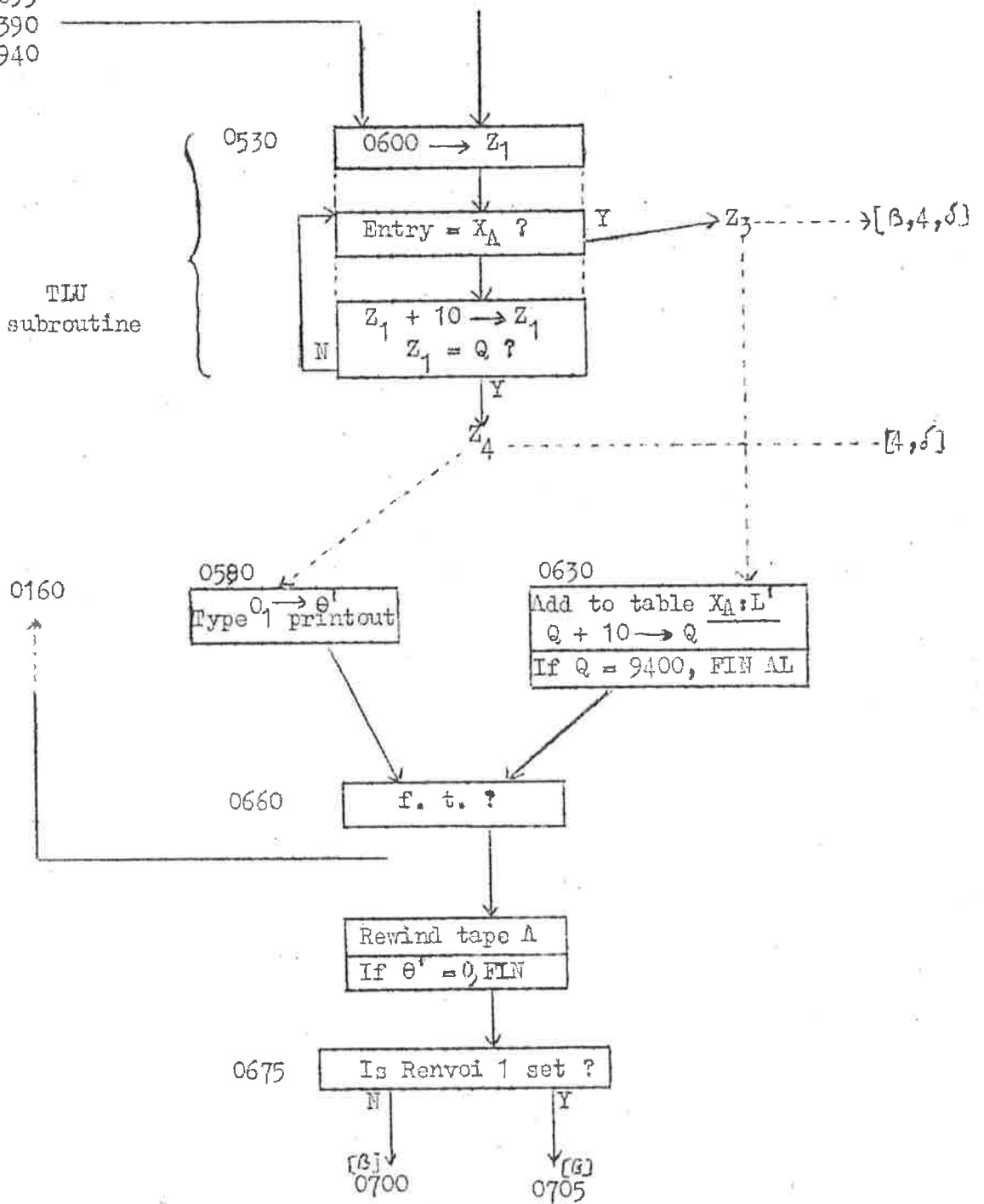
15. Block Diagrams

Block α



[B] 1095
[4] 1390
[δ] 1940

TLU
subroutine



Block β

[α] 0675

0700

Serial 0002 $\rightarrow$ C

0705

Set L, 1, T_1, T_2, U_1, U_2
 θ, P, P_N
 READ $\Delta, \rightarrow F$
 $X_P = 38 ?$

N

0760

TRANSFER 0015 - 0090
 from order drum $\rightarrow$ C

[p] 2135

2485

[j] 2010

0800

READ $\Delta, \rightarrow F$

0805

Store $X, X_P, X_{P2}, X_A, X_I, X_S$

$X_P = 39 ?$

Y

[δ] 1900

$X_P = 05 ?$

Y

[ϵ] 1915

$X \rightarrow U_1$

$89 \geq X_P \geq 70 ?$

Y

[7] 1130

$49 \geq X_P \geq 40 ?$

Y

[4] 1280

$X_S = 8 ?$

Y

N

1090

$1100 \rightarrow Z_3, 1395 \rightarrow Z_4$

$\rightarrow$ (TLU) 0530

1100

Replace X_A
 Change U_1

$\leftarrow$ (TLU) 0550
 [4] 1445

0965

$X_S \geq 90 ?$

Y

[4] 1500

$X_S \geq 50 ?$

Y

[5] 1210

If $X_{P2} \geq 5$, subtract 5;
 result $\rightarrow F$.

$F = 0 ?$

Y

[2] 1520

$X_P \leq 09 ?$

Y

[0] 1730

$X_P \leq 19 ?$

Y

[1] 1660

$X_P \geq 36 ?$

N

1045

Shift X_P r. 4;
 $+X_A + X_I \rightarrow T_1$

1065

Shift F r. 5;
 $+ .00004 + X_A$
 $+ X_I \rightarrow T_1$

[p] 2070

Block 7

[B]0925

[P]2475

1130

0 → 0. 1 = 0 ?

0 → T₂, -1 → 0, 0 → Z₄

[P]2160

1140

Fill T₁, U₁. 5 → 1.
Shift X_P-70 r. 1; → Z₁
X_P = 70? (40 → X_P)

Y

[4]1365

N

1170

Use Z₁ to look up tr.
address; fill T₂, U₂

[P]2070

Block 5

[S]0980

1210

Use X_P-50 to look up tr.
prefix. Prefix neg. ?

Y

N

If X_I ≥ 5, subtract 5;
leave in g.c.

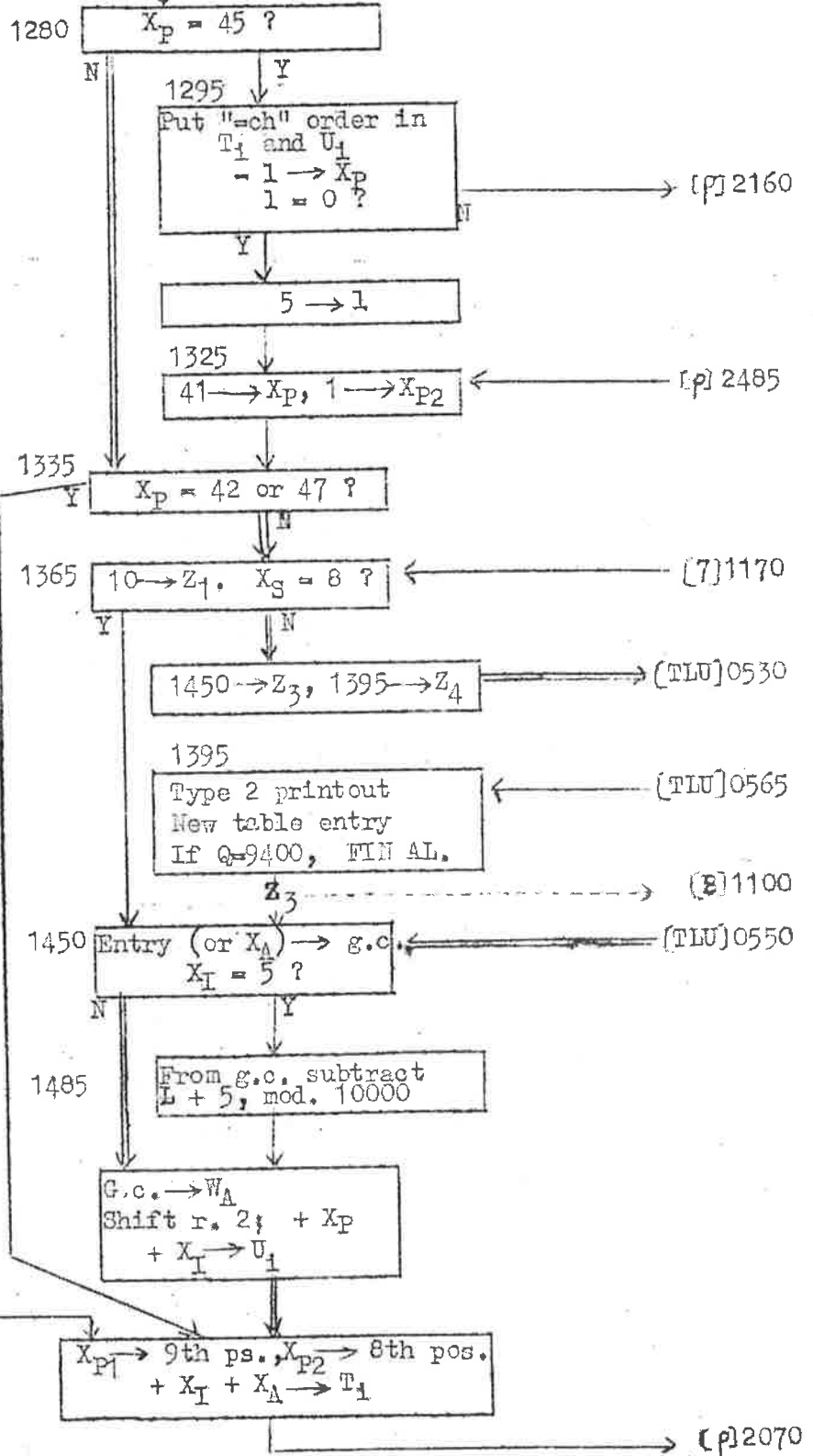
1260

+ X_A + prefix → T₁

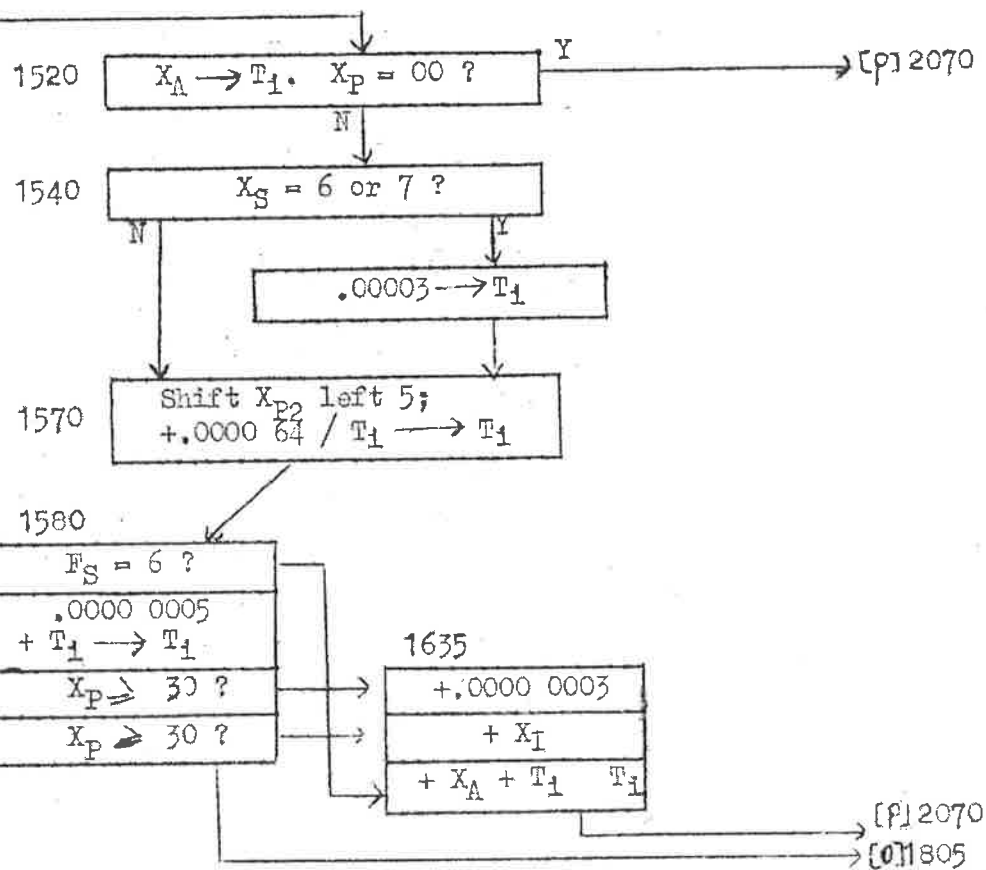
[P]2070

Block 4

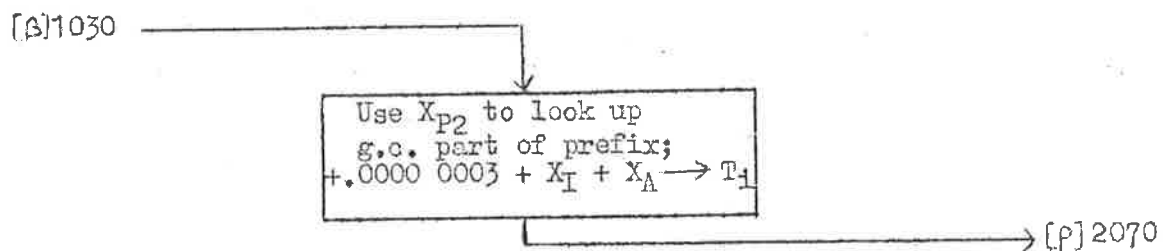
[B]0945



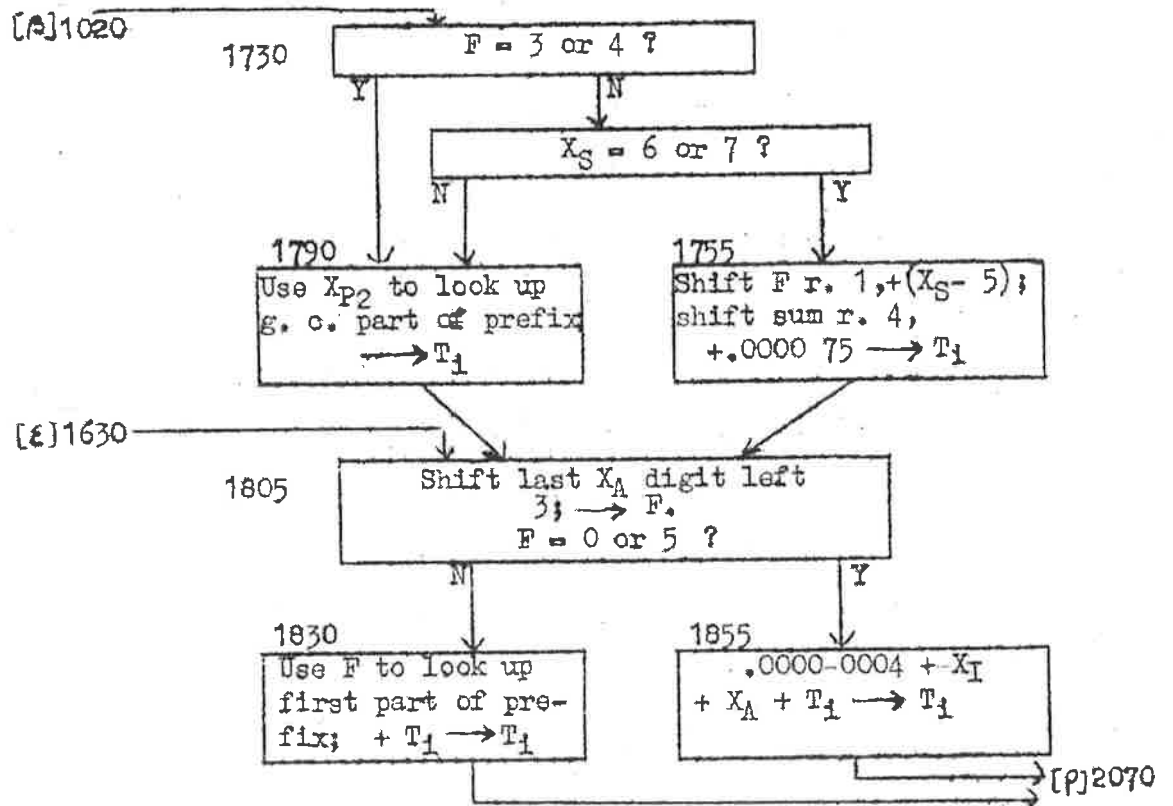
Block 2
[B]1010



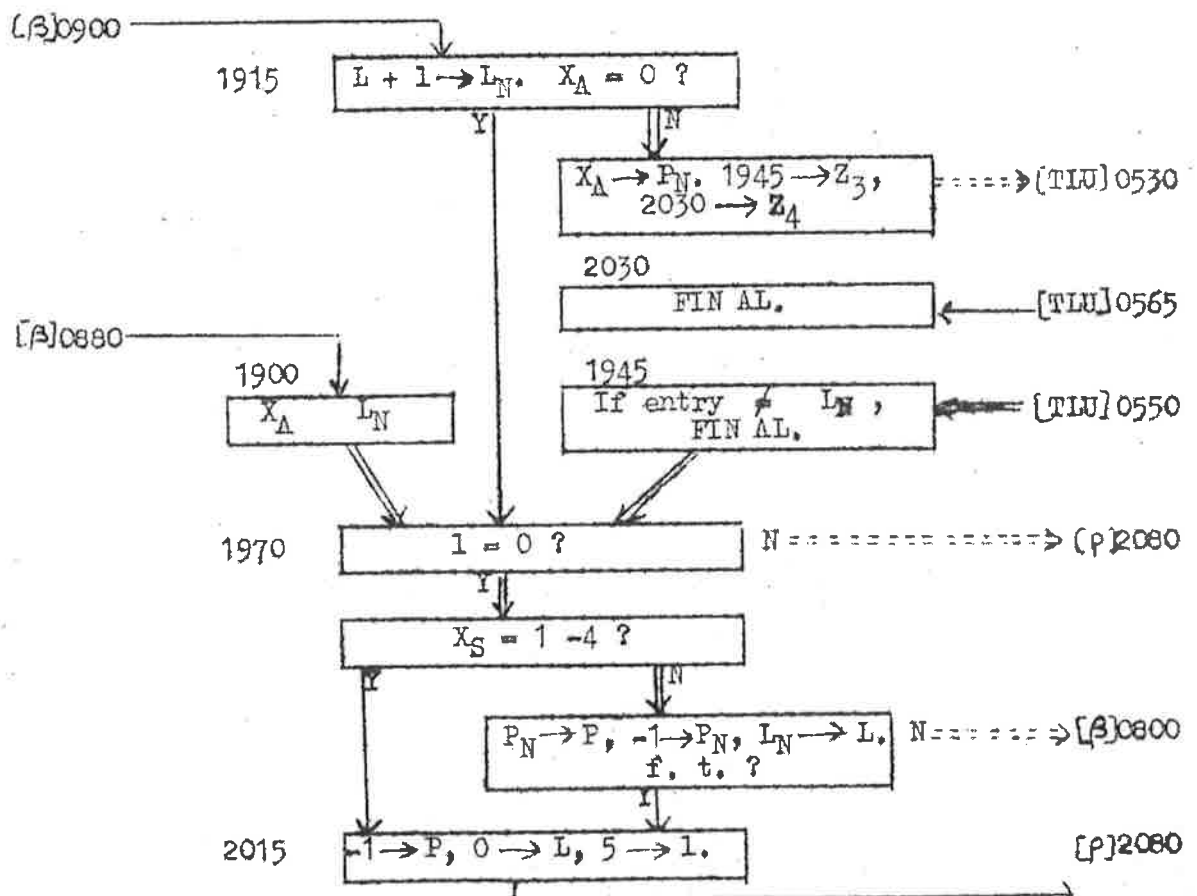
Block 1

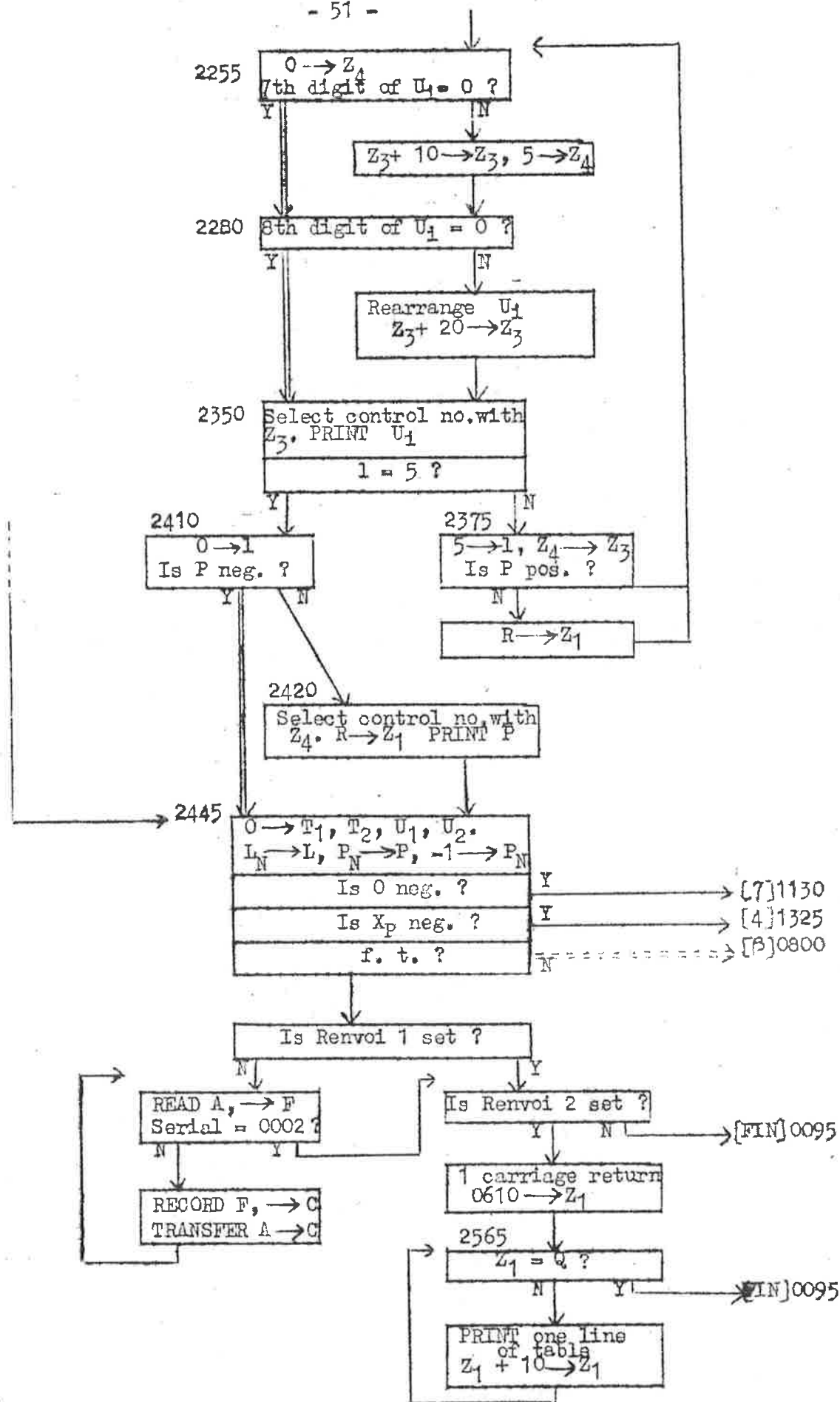


Block 0



Block 6





Chapter IV : Other Routines

1. Pseudocode Printing Routine

Included on the Pseudocode Translation Routine tape are two other shorter routines. The first of these is described in the present section, and the second in § 2.

The Pseudocode Printing (PCP) Routine is used for printing pseudoprogram tapes so that they can be verified before being translated. On the PCT tape, the PCP coding comes directly after the PCT instructions. There is a blank word with serial 7940, and then the words of the PCP routine with serial 7945-8085. The routine has been given these high numbers with the hope that it will remain undisturbed on the order drum most of the time. (The number drum is not used.) However, the routine is coded relatively and so can be placed in any set of 29 consecutive locations on the order drum. It can equally well be placed on the number drum if, in the following order word :

7950 92000 0000 90000 0005

the second address is changed from 0005 to 0008.

The PCP routine prints pseudo-orders in a single column. Each order is preceded by its serial number on the tape. The prefix and address are separated by a space, and the address and index digit by another space. However, the eighth digit is printed only when it is nonzero and the index digit is printed only when it is nonzero or when the eighth digit is printed. Each pseudo-order is printed in approximately two-thirds of the time (five seconds) required to print a complete number word.

If desired, the PCP routine prints the second tranche of the pseudo-program tape in the usual "Prog. Nombres" format.

1.1. Use of the routine

The printer key should be set to "Progr. Nombres". The length of a line may be 18, 19, or 20 spaces, so it is necessary to short-circuit the end-of-line check contact (on the back of the typewriter).

Register OB must be filled by hand, putting the tape to be printed in position A. Renvoi 1 should be set to "Inhiber" if it is desired to print the tape's second tranche (in the usual "Prog. Nombres" format), and set to "Autor," otherwise.

The routine starts at order 7950. It ends at 7945, so in order to repeat the program it is necessary only to press "Programme Start".

The printing normally starts with the first word on the tape; if it is desired to start at some other point, the tape should be set to the proper point, "Org. Tambours" set to 5, and the routine started at order 7955. The printing of pseudo-orders normally stops at the end of the first tranche; if it is desired to halt at some specific serial number, that number should be placed in S, and in the following order word :

7965 48500 0000 06900 0000

the prefix 48500 should be changed to 43500.

1.2. Details of the routine

The PGP routine orders occupy locations 7945-8060 and the constants 8065-8085, as shown below.

Note that the constant in location 8080 is used twice : first, in order 7980, in an "alteration tri;" second, in order 8015, to achieve a left shift of eight. The constant 8085 is used only in order 8040, to shift the pseudo order three places to the left just before it is printed.

Coding					Notes
7940	00000	0000	00000	0000	
7945	42000	0000	00000	0000	FIN
7950	92000	0000	90000	0005	Rewind A. Org. drums
7955	95000	0009	01700	0105	Org. transfer : one word, 8065 $\rightarrow$ Z ₂ 1 $\rightarrow$ Z ₄ . READ A, $\rightarrow$ F. If f. t., 0 $\rightarrow$ Z ₄ . V $\rightarrow$ Z ₁ .
7960	06900	0001	52000	0000	
7965	48500	0000	06900	0000	
7970	02566	0000	06646	0000	
7975	00021	0000	02066	0000	7th, 8th digits of F $\rightarrow$ E
7980	00043	0000	04536	0095	
7985	07146	0000	03035	0000	Shift 7th, 8th digits of F right 1 in F. If both digits = 0, Rv 8030
7990	02166	0000	07496	0000	
7995	03036	1000	02176	0000	
8000	02066	0000	07046	0000	
8005	41500	0020	01200	0005	Z ₂ + 5 $\rightarrow$ Z ₂ . If 8th digit = 0, Rv 8030
8010	02166	0000	03036	1000	
8015	00033	0000	04536	0060	
8020	03076	0001	07446	0000	
8025	41500	0000	01200	0005	Z ₂ + 5 $\rightarrow$ Z ₂
8030	00011	0000	04266	0000	Select control number with Z ₂ , $\rightarrow$ E. PRINT F, with serial Z ₁ .
8035	07146	0000	02066	0000	
8040	04566	0040	05146	0000	
8045	03476	0000	07446	0000	If Z ₄ = 1, Rv 7955
8050	41500	9900	44500	9890	If Renvid 1 is set, Rv 7945
8055	95000	0000	50000	00000	PRINT second tranche of A. Rv 7945.
8060	40500	9880	00000	0000	
8065	-.44400	00004	44444	+44	
8070	-.44400	00004	04444	+44	
8075	-.44400	00004	00444	+44	
8080	+.00000	01100	00000	+00	
8085	+.00000	00000	00000	+03	

2. Read-Off-Drum Routine

The second short routine included on the PCT tape is the Read-Off-Drum (ROD) Routine. This routine has no direct connection with pseudocoding, but is included because of its general usefulness. It is used for transferring a consecutive set of words from either drum to the printer or to magnetic tape.

On the PCT tape, the ROD coding comes directly after the PCP coding. There is a blank word with serial 8090, next the orders of the ROD routine with serials 8095-8160, and finally a blank word with serial 8168 which ends the first tranche and group of the PCT tape. The ROD routine has been given these high serial numbers with the hope that it will remain undisturbed on the order drum most of the time, like the permanent subroutines. However, the routine is coded relatively and so can be placed in any set of 14 consecutive locations on the order drum or number drum.

The ROD routine is particularly useful for two purposes. The first is to print out a part of the contents of the drums in order to see whether some program is proceeding correctly. The second is, when corrections have been made to a program stored in the computer, to produce a new tape containing the corrected program for future use.

2.1. Use of the routine

If the printer is to be used, the format key should be to "Prog. Ordres" or "Prog. Nombres" as desired. If a tape is to be used, register OB must be filled so as to put the desired tape in position C.

"Org. Tambours" must be set to 5 or 6, according as the reading is to be from the order drum or number drum. Index registers Z_1 , Z_2 , Z_3 must be filled as follows : the first and last location wanted read off the drum are placed in Z_1 and Z_2 respectively. If the words are to be printed, Z_3 should be set equal to zero. If the words are to be recorded on tape, Z_3 should be unequal to zero.

The routine starts at 8100. It ends at 8095, so that in order to repeat it, one need only reset Z_1 , Z_2 , Z_3 and press "Programme Start".

It may be noted that in addition to the registers mentioned above, the routine destroys the previous contents of register E, the check register, and the groupe calculateur.